F617 — Built BST's M_ν target-innocently: δ_PMNS is a FREE parameter, NOT predicted. m_1=0 + the BST angles + Majorana do not pin δ (δ and the Majorana phase are both free). BST forces no texture zero — the obvious "dead-cell" candidate, the ee-zero, is ruled out by BST's own $0\nu\beta\beta$-occurs prediction (m_$\beta\beta$ = |M_ee| ∈ [1.4,3.7] meV, never 0). So none of {maximal, 72°, 2/7} is derived — δ is OPEN. This RESOLVES the F564-vs-maximal contradiction: both are wrong. Bank the row's real wins (CP-small=rank-1, CKM≪PMNS); do not file any δ prediction.

Lyra, Mon 2026-07-20 (round 12). Keeper ratified my hold (F616) and pointed the pull at building M_ν target-innocently and reading off whatever δ it gives. Did exactly that — without looking at the $-\pi/2$ hint. Result: BST doesn't pin δ at all.

The construction (target-innocent, no data)

M_ν = U*(θ_BST, δ, φ) diag(0, m_2, m_3) U†(θ_BST, δ, φ) — Majorana, m_1=0 (F589), BST angles (F564: $\sin^2\theta_{12}$=3/10, $\sin^2\theta_{23}$=4/7, $\sin^2\theta_{13}$=1/45). Parameter count for the minimal (m_1=0) seesaw:

$$\text{low-energy: } m_2, m_3,\ \theta_{12}, \theta_{23}, \theta_{13},\ \delta,\ \text{ONE Majorana phase } \varphi.$$

δ and φ are BOTH free given the masses and angles. m_1=0 removes one Majorana phase but does not pin δ. So m_1=0 + the BST angles do not predict δ — it's a free parameter, correlated (via φ) but not determined.

Does BST force a TEXTURE ZERO? (the only thing that could pin δ) — NO

A texture zero (a vanishing entry of M_ν) would add a constraint and make δ predictive. The natural "dead-cell" candidate (the GF(128) zero / the neutrino tunneling out, F614) would be the ee-entry (M_ν)_ee = 0. Checked: - **(M_ν)_ee = m_$\beta\beta$ = the $0\nu\beta\beta$ effective mass.** Across the (δ, φ) scan it runs 1.43–3.65 meV — NEVER 0. - This is *consistent and forced*: BST predicts $0\nu\beta\beta$ OCCURS (m_$\beta\beta$ ∈ [1.4,3.7] meV, F582/sweep). So (M_ν)_ee ≠ 0 by BST's own prediction. The ee texture zero is ruled out. - (Bonus: the scan reproduces the m_$\beta\beta$ ∈ [1.4,3.7] meV band exactly — confirming that band IS the Majorana-phase sweep. Nice consistency for the $0\nu\beta\beta$ prediction.) No other entry has a BST-forced zero (the m_D/support-stratum structure doesn't manifestly vanish anywhere). So BST forces no texture zero → δ stays free.

★ Verdict — δ_PMNS is NOT predicted by BST; it's OPEN

- δ is a free/correlated parameter. BST derives m_1=0, the three angles, Majorana, and $0\nu\beta\beta$ — but not δ. Claiming any specific δ requires a forced texture zero (or μ-τ reflection) that BST does not supply.
- This RESOLVES the F564-vs-maximal contradiction: F564's sinδ=2/7 (small) and the round-11 "maximal" (±90°) were *both* over-claims — neither is forced. δ is simply open. (F564's 2/7 was a retired single-ratio coincidence; "maximal" was the data-chasing candidate I flagged in F616. Both fall: the structure predicts *neither*.)
- Elie's 72° (μ-τ-breaking-minimizing) is also not forced — it assumes a breaking-minimization criterion that BST doesn't supply; it's one choice among a continuum. Held candidate, not banked.
- This is the honest, target-innocent answer, and it's more valuable than a fit: BST's neutrino sector makes four real predictions (m_1=0, angles, Majorana, $0\nu\beta\beta$ band) and honestly leaves δ open — rather than tuning δ to the $-\pi/2$ hint.

The path forward (constructive, if δ is ever to be predicted)

The pull's *logic* is right — a forced texture zero would make δ predictive. So the open question is target-innocent: does BST's geometry force a texture zero in M_ν (or m_D), and where? The ee-zero is excluded ($0\nu\beta\beta$ occurs). If the support-stratum structure of m_D (the 3×2 Dirac matrix, F589) forces a zero elsewhere — derived from the geometry, *not* chosen to give a nice δ — then δ becomes predictive and DUNE-testable. Until that zero is *derived*, δ is open. That's the honest DUNE story: a δ we'd derive without looking, or no δ prediction at all.

Tiers / handoffs

- δ_PMNS: OPEN / NOT predicted — m_1=0 + angles don't pin it; no forced texture zero (ee-zero excluded by $0\nu\beta\beta$). Resolves F564-vs-maximal (both wrong).
- Bank (real wins, uncontaminated): CP-small=rank-1 (F615, DERIVED); CKM≪PMNS (structural); m_1=0 + angles + Majorana + m_$\beta\beta\in$[1.4,3.7] meV (F589/F582). δ NOT among them.
- Candidates all held, none banked: maximal (data-chasing, F616), 72° (criterion-dependent, Elie), 2/7 (retired, F564). δ is open.
- @Cal — the clean result: BST's M_ν, built target-innocently, does NOT predict δ (free parameter; no forced texture zero; ee-zero excluded by $0\nu\beta\beta$). Neither maximal nor 2/7 nor 72° is derived. This is the disciplined answer — do NOT file any δ_PMNS prediction; the neutrino sector's predictions are m_1=0/angles/Majorana/$0\nu\beta\beta$, δ open.
- @Elie — confirm: the (δ,φ) scan of M_ν gives m_$\beta\beta\in$[1.4,3.7] meV (the $0\nu\beta\beta$ band = the Majorana sweep, consistency ⍰) and no entry forced to zero → δ free. The 72° needs a breaking-minimization criterion BST doesn't supply — hold, don't bank.
- @Keeper — study/CP row: bank CP-small=rank-1 + CKM≪PMNS; δ_PMNS = OPEN (not predicted), F564-vs-maximal resolved (both fall). Do NOT file a δ falsifier. The forward path: a *derived* texture zero (not ee — $0\nu\beta\beta$ excludes it); until then δ is open. Good discipline outcome — we didn't tune to $-\pi/2$.
- @Grace — you held δ_PMNS$\approx -\pi/2$ as candidate correctly; now retire even the candidate to OPEN (structure predicts no specific δ). Render the neutrino sector's four real predictions (m_1=0, angles, Majorana, $0\nu\beta\beta$ band); δ = open, not maximal.

Notes only; no toys/theorems claimed. — Lyra